Statistics, Counting & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 5 red and 3 blue balls. Find the probability that 3 balls drawn without replacement are all red.

Method 1 (Sequential Conditional Probability): Total balls = $5 + 3 = 8$. 1st ball red: $5/8$ 2nd ball red: $4/7$ 3rd ball red: $3/6 = 1/2$ $P = \frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} = \frac{60}{336} = \mathbf{\frac{5}{28}} \approx 0.1786$ Method 2 (Combinations): $P = \frac{\binom{5}{3}}{\binom{8}{3}} = \frac{10}{56} = \mathbf{\frac{5}{28}}$

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $P(A) = 0.6$ and $P(B|A) = 0.4$, find the joint probability $P(A \cap B)$.

Apply the conditional probability multiplication rule $P(A \cap B) = P(B \mid A) \cdot P(A)$: $P(A \cap B) = 0.4 \times 0.6 = \mathbf{0.24}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 3 red, 4 white, and 5 blue cards. Find the probability of drawing a card that is not white.

Step 1: Total cards: $3 + 4 + 5 = 12$. Step 2: Non-white cards: $3 \text{ red} + 5 \text{ blue} = 8$ (or $12 - 4 = 8$). Step 3: Probability: $P(\text{not white}) = \frac{8}{12} = \mathbf{\frac{2}{3}} \approx 0.667$

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple (LCM) of 12 and 18.

Step 1 (Prime Factorizations): $12 = 2^2 \times 3^1$ $18 = 2^1 \times 3^2$ Step 2 (Highest Powers): Take the maximum power of each prime factor: $\text{lcm}(12, 18) = 2^{\max(2,1)} \times 3^{\max(1,2)} = 2^2 \times 3^2 = 4 \times 9 = \mathbf{36}$ Check via Duality: $\text{lcm}(a, b) = \frac{a \cdot b}{\gcd(a, b)} = \frac{12 \times 18}{6} = 36$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical asymptote(s) of $f(x) = \frac{3x}{x^2 - 4}$.

Step 1: Vertical asymptotes occur where the denominator is zero while the numerator is non-zero. $x^2 - 4 = 0 \Rightarrow (x - 2)(x + 2) = 0 \Rightarrow x = 2$ and $x = -2$ Step 2: Verify numerator at these values: $3(2) = 6 eq 0$ and $3(-2) = -6 eq 0$. The vertical asymptotes are the lines $\mathbf{x = 2}$ and $\mathbf{x = -2}$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\lim_{x\to\infty} \frac{3x^2+1}{x^2-2}$.

Step 1: Divide numerator and denominator by the highest degree term x^2: $\lim_{x\to\infty} \frac{3x^2+1}{x^2-2} = \lim_{x\to\infty} \frac{3+\frac{1}{x^2}}{1-\frac{2}{x^2}}$ Step 2: As $x\to\infty$, $\frac{1}{x^2}\to 0$ and $\frac{2}{x^2}\to 0$: $\frac{3+0}{1-0} = \mathbf{3}$

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set has standard deviation 3 and mean 20. If every value in the data set is increased by 5, find the new mean and new standard deviation.

New Mean: Adding a constant $c = 5$ shifts the mean by $+5$: $\text{New Mean} = 20 + 5 = \mathbf{25}$ New Standard Deviation: Standard deviation measures dispersion relative to the mean; adding a constant shifts all points uniformly without altering spread: $\text{New Standard Deviation} = \mathbf{3}$ (unchanged)

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the finite series $\sum_{k=1}^{4} (2k-1)$.

Step 1: Expand the summation terms for $k = 1, 2, 3, 4$: $k=1: 2(1)-1=1$ $k=2: 2(2)-1=3$ $k=3: 2(3)-1=5$ $k=4: 2(4)-1=7$ Step 2: Add the terms: $1+3+5+7 = \mathbf{16}$ (Property: The sum of the first n odd positive integers is $n^2 = 4^2 = 16$).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Expand $(a+b)^5$ and find the coefficient of the term a^2b^3.

By the Binomial Theorem, the general term in $(a+b)^n$ is $T = \binom{n}{k} a^{n-k}b^k$. For a^2b^3, $n=5$ and $k=3$: $\text{Coefficient} = \binom{5}{3} = \frac{5!}{3!\cdot 2!} = \frac{5\times 4}{2\times 1} = \mathbf{10}$

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.

Step 1: Total sum of the original 10 numbers: $S_{10} = 10\times 15 = 150$ Step 2: Total sum of the remaining 9 numbers: $S_9 = 9\times 14 = 126$ Step 3: The removed value is: $x = S_{10} - S_9 = 150 - 126 = \mathbf{24}$

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the amplitude and period of $y = -4\cos\left(\frac{\pi x}{3}\right)$.

Amplitude: $|A| = |-4| = \mathbf{4}$. Period: For $y = A\cos(Bx)$ with $B = \frac{\pi}{3}$: $T = \frac{2\pi}{|B|} = \frac{2\pi}{\pi/3} = 2\pi\times\frac{3}{\pi} = \mathbf{6}$

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance from $(1,\ -2)$ to $(5,\ 1)$.

Apply the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:
$d = \sqrt{(5-1)^2 + (1-(-2))^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = \mathbf{5}$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars? Show your complete algebraic work.

Final Answer: 8. First, calculate the current sum of the ratings from the bar graph: $\text{Sum} = (1\times 2) + (2\times 6) + (3\times 6) + (4\times 4) + (5\times 2) = 2 + 12 + 18 + 16 + 10 = 58$ The current number of customers is $2 + 6 + 6 + 4 + 2 = 20$. Let x represent the number of additional 5-star ratings. Adding x ratings of 5 stars increases the total sum of ratings by $5x$ and increases the total number of customers to $20 + x$. Set up the equation for the new mean: $\frac{58 + 5x}{20 + x} = 3.5$ Multiply both sides by $20 + x$: $58 + 5x = 3.5(20 + x)$ $58 + 5x = 70 + 3.5x$ Subtract $3.5x$ and 58 from both sides: $1.5x = 12$ $x = \frac{12}{1.5} = 8$ Thus, 8 additional 5-star ratings are required. A common misconception is to compute the point deficit $3.5 \times 20 - 58 = 12$ and assume 12 reviews are needed, which forgets that the denominator increases with every new review.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music? Show your complete algebraic work.

Final Answer: $\frac{7}{20}$. To find the probability that a randomly chosen student is enrolled in Music, divide the number of students enrolled in Music by the total number of students enrolled across all four courses. 1. Find the total number of students: $\text{Total} = 8 + 14 + 6 + 12 = 40$ 2. Find the number of students in Music: $n(\text{Music}) = 14$ 3. Calculate the probability: $P(\text{Music}) = \frac{14}{40} = \frac{7}{20}$ Common misconceptions: - $\frac{7}{13}$ computes the odds in favor (ratio of students in Music to students not in Music, $\frac{14}{26}$) rather than probability. - $\frac{13}{20}$ represents the complement probability (the probability that the student is NOT enrolled in Music, $1 - \frac{7}{20} = \frac{13}{20}$). - $\frac{7}{50}$ incorrectly assumes the total is 100 students.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Advanced Number Theory

The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama? Show your complete algebraic work.

Final Answer: $\frac{4}{25}$. First, find the total number of students enrolled in the four classes by adding the frequencies from the bar graph: $\text{Total} = 12 + 16 + 8 + 14 = 50$ From the graph, the number of students enrolled in Drama is 8. The probability of selecting a student enrolled in Drama is the ratio of the number of Drama students to the total number of students: $P(\text{Drama}) = \frac{8}{50} = \frac{4}{25}$ - $\frac{1}{4}$ incorrectly assumes each elective has an equal number of students. - $\frac{4}{21}$ incorrectly divides Drama students by the remaining students ($8/42$) instead of the total. - $\frac{7}{25}$ is the probability of selecting a student in French ($14/50$).

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Advanced Algebra & Systems

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days? Show your complete algebraic work.

Final Answer: 6.0. To find the mean number of cars sold per day, sum the number of cars sold across all 4 days and divide by the total number of days (4): $\text{Total cars sold} = 5 + 9 + 4 + 6 = 24$ $\text{Mean} = \frac{24}{4} = 6.0$ - The distractor 5.5 is the median of the data set (ordered: 4, 5, 6, 9; median = $\frac{5+6}{2} = 5.5$). - The distractor 6.5 is the midrange ($\frac{4+9}{2} = 6.5$). - The distractor 8.0 results from incorrectly dividing the sum by 3 instead of 4.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Functions

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse? Show your complete algebraic work.

Final Answer: $\frac{6}{11}$. To find the probability that a randomly chosen triangle is obtuse, we determine the total number of possible triangles and the number of obtuse triangles. **Step 1: Total number of triangles** Choosing any 3 distinct vertices from the 12 vertices of the regular 12-gon gives: $\binom{12}{3} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$ **Step 2: Counting right and acute triangles** - **Right Triangles:** By Thales' Theorem, an inscribed triangle is a right triangle if and only if one of its sides is a diameter of the circumscribed circle. A regular 12-gon has $\frac{12}{2} = 6$ diameters. For each diameter, choosing any of the remaining $12 - 2 = 10$ vertices creates a right triangle. Since a triangle cannot contain more than one right angle, each right triangle is counted exactly once: $N_{\text{right}} = 6 \times 10 = 60$ - **Acute Triangles:** An inscribed triangle is acute if and only if the center of the circle lies strictly in its interior. Equivalently, the three arc lengths a, b, c between consecutive chosen vertices (measured in units of $\frac{1}{12}$ of the circle) must satisfy $a + b + c = 12$ with $1 \le a, b, c \le 5$. - Partition $\{4, 4, 4\}$: equilateral triangles. There are $\frac{12}{3} = 4$ such triangles. - Partition $\{5, 5, 2\}$: isosceles acute triangles. There are 3 cyclic permutations, giving $\frac{12 \times 3}{3} = 12$ triangles. - Partition $\{5, 4, 3\}$: scalene acute triangles. There are $3! = 6$ cyclic permutations, giving $\frac{12 \times 6}{3} = 24$ triangles. Thus, the total number of acute triangles is: $N_{\text{acute}} = 4 + 12 + 24 = 40$ **Step 3: Counting obtuse triangles** Subtracting the right and acute triangles from the total: $N_{\text{obtuse}} = 220 - 60 - 40 = 120$ *(Alternative direct count)*: A triangle is obtuse if and only if its longest subtended arc L strictly exceeds a semicircle, so $L \in \{7, 8, 9, 10\}$. For a fixed side of arc length L, the third vertex must lie strictly inside the remaining arc of length $12 - L$, which offers $11 - L$ internal vertices. Summing over all 12 positions for each L: $12 \times [(11 - 7) + (11 - 8) + (11 - 9) + (11 - 10)] = 12 \times (4 + 3 + 2 + 1) = 120$ **Step 4: Calculating the probability** $P(\text{obtuse}) = \frac{120}{220} = \frac{6}{11}$ **Misconception Analysis:** - Choice A, $\frac{3}{11} = \frac{60}{220}$, is the probability of selecting a right triangle. - Choice B, $\frac{5}{11}$, typically arises from neglecting one class of obtuse triangles or calculating $1 - P(\text{obtuse})$. - Choice D, $\frac{7}{11}$, results from mistakenly grouping right triangles together with obtuse triangles.

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18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 4.01 Advanced Trigonometry**

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee? Show your complete algebraic work.

Final Answer: $\frac{3}{7}$. To find the conditional probability, we determine the number of committee selections that satisfy the given condition (the reduced sample space) and the number of those selections that contain exactly 2 mathematicians. 1. **Analyze the condition:** The committee must have 4 members chosen from 3 professions with at least one member from each profession. By the Pigeonhole Principle, the members must be partitioned as $2 + 1 + 1$. Thus, exactly one profession will have 2 representatives, and the remaining two professions will each have 1 representative. 2. **Count the outcomes for each mutually exclusive composition:** - **Case 1: 2 mathematicians, 1 physicist, 1 engineer** $\binom{4}{2} \times \binom{3}{1} \times \binom{3}{1} = 6 \times 3 \times 3 = 54$ - **Case 2: 1 mathematician, 2 physicists, 1 engineer** $\binom{4}{1} \times \binom{3}{2} \times \binom{3}{1} = 4 \times 3 \times 3 = 36$ - **Case 3: 1 mathematician, 1 physicist, 2 engineers** $\binom{4}{1} \times \binom{3}{1} \times \binom{3}{2} = 4 \times 3 \times 3 = 36$ 3. **Calculate the total number of outcomes satisfying the condition:** $N(\text{condition}) = 54 + 36 + 36 = 126$ 4. **Compute the conditional probability:** The number of favorable outcomes (where exactly 2 mathematicians are chosen) is 54.
$P(2 \text{ mathematicians} \mid \text{at least 1 of each}) = \frac{54}{126} = \frac{3}{7}$ **Common Misconceptions:** - $\frac{1}{3}$ assumes all three professions are equally likely to have 2 members, failing to account for the larger pool of mathematicians (4 candidates vs. 3). - $\frac{9}{35}$ is the unconditional probability $\frac{54}{\binom{10}{4}} = \frac{54}{210}$, forgetting to restrict the sample space to the given condition. - $\frac{3}{5}$ represents the overall probability of the condition occurring, $\frac{126}{210}$, rather than the required conditional probability.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Pre-Calculus Concepts

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3:5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally? Show your complete algebraic work.

Final Answer: 15. 1. Find the total number of green blocks after the addition: Since the new total is 60 and the probability of selecting a green block is $\frac{1}{3}$, the number of green blocks is: $60 \times \frac{1}{3} = 20$ 2. Determine the number of red and blue blocks: The remaining blocks in the bin are red and blue. Since green blocks account for 20 of the 60 blocks, the combined total of red and blue blocks is: $60 - 20 = 40$ (Alternatively, originally there were $20 - 12 = 8$ green blocks, and the original total was $60 - 12 = 48$ blocks, leaving $48 - 8 = 40$ red and blue blocks.) 3. Use the given ratio to find the number of red blocks: The ratio of red blocks to blue blocks is $3:5$, so red blocks represent $\frac{3}{3+5} = \frac{3}{8}$ of the red-and-blue total: $\text{Red blocks} = \frac{3}{8} \times 40 = 15$ Common Pitfall: Subtracting only the 12 added green blocks from 60 gives 48. Assuming there were no green blocks originally leads to $\frac{3}{8} \times 48 = 18$, which incorrectly ignores the green blocks that were already present.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Of 31 students, 13 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball? Show your complete algebraic work.

Final Answer: $\frac{11}{13}$. Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{11}{13} = \frac{11}{13}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A charity raffle sells $1,000$ tickets at 5 each. There is one grand prize of $1,000$ and two second prizes of 500 each. What is the expected net gain for a person purchasing one ticket?

a. -3.00 ✓

b. -2.00

c. -1.50

d. 0.00

Expected payout $= \frac{1}{1000}(1000) + \frac{2}{1000}(500) = 1 + 1 = 2$. Net expected gain $= 2 - 5 = -3$ dollars.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random variable X has probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. What is the expected value $E(X)$?

a. 2.0

b. 2.1 ✓

c. 2.2

d. 2.3

$E(X) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

a. 3

b. 5

c. 9 ✓

d. 16

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. Approximately what percentage of test takers score between 400 and 600?

a. 50%

b. 68% ✓

c. 95%

d. 99.7%

Scores between 400 and 600 lie within 1 standard deviation of the mean ($\mu \pm \sigma$). By the Empirical Rule, this encompasses approximately 68% of data.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scores 84 on a test with mean 72 and standard deviation 8. What is the student's z-score?

a. 1.25

b. 1.50 ✓

c. 1.75

d. 2.00

Formula: $z = \frac{x - \mu}{\sigma} = \frac{84 - 72}{8} = \frac{12}{8} = 1.50$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a normal distribution with mean 50 and standard deviation 5, what value corresponds to a z-score of -1.6?

a. 42 ✓

b. 44

c. 46

d. 48

$x = \mu + z\sigma = 50 + (-1.6)(5) = 50 - 8 = 42$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

a. 0.50

b. 0.60 ✓

c. 0.75

d. 0.30

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If two events A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?

a. 0.70

b. 0.80 ✓

c. 0.90

d. 0.1575

For mutually exclusive events, $P(A \cap B) = 0$. Therefore, $P(A \cup B) = P(A) + P(B) = 0.35 + 0.45 = 0.80$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

a. 0.019 ✓

b. 0.020

c. 0.095

d. 0.95

$P(\text{Disease} \cap \text{Positive}) = P(\text{Disease}) \times P(\text{Positive} \mid \text{Disease}) = 0.02 \times 0.95 = 0.019$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many ways can 6 students be seated in a circle?

a. 120 ✓

b. 720

c. 60

d. 24

Circular permutations of n distinct objects is $(n-1)!$. Here $(6-1)! = 5! = 120$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

a. 2

b. 3 ✓

c. 4

d. 11

There are 2 colors (pigeonholes). By the Pigeonhole Principle, drawing $2 + 1 = 3$ socks guarantees at least two socks of the same color.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

a. $50,400$ ✓

b. $100,800$

c. $302,400$

d. $3,628,800$

Word length is 10 letters: 3 S's, 3 T's, 2 I's, 1 A, 1 C. Permutations $= \frac{10!}{3!\,3!\,2!} = \frac{3,628,800}{6 \times 6 \times 2} = \frac{3,628,800}{72} = 50,400$.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 32 students, 21 play soccer and 10 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{21}{32}$

b. $\frac{10}{11}$

c. $\frac{10}{21}$ ✓

d. $\frac{11}{21}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{10}{21} = \frac{10}{21}$.

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 39 students, 19 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{8}{19}$

b. $\frac{11}{20}$

c. $\frac{11}{19}$ ✓

d. $\frac{11}{39}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{11}{19} = \frac{11}{19}$.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 27 students, 17 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{3}{10}$

b. $\frac{14}{17}$

c. $\frac{3}{17}$ ✓

d. $\frac{17}{27}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{3}{17} = \frac{3}{17}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 37 students, 29 play soccer and 4 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{25}{29}$

b. $\frac{4}{29}$ ✓

c. $\frac{1}{2}$

d. $\frac{4}{37}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{4}{29} = \frac{4}{29}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 36 students, 10 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{5}{18}$

b. $\frac{1}{12}$

c. $\frac{7}{10}$

d. $\frac{3}{10}$ ✓

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{3}{10} = \frac{3}{10}$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

Of 40 students, 10 play soccer and 7 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{7}{30}$

b. $\frac{7}{10}$ ✓

c. $\frac{3}{10}$

d. $\frac{1}{4}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{7}{10} = \frac{7}{10}$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Of 35 students, 16 play soccer and 9 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{9}{19}$

b. $\frac{9}{16}$ ✓

c. $\frac{9}{35}$

d. $\frac{16}{35}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{9}{16} = \frac{9}{16}$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Of 34 students, 17 play soccer and 2 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{1}{17}$

b. $\frac{2}{17}$ ✓

c. $\frac{15}{17}$

d. $\frac{1}{2}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{2}{17} = \frac{2}{17}$.

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41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

Of 40 students, 17 play soccer and 5 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{12}{17}$

b. $\frac{5}{23}$

c. $\frac{1}{8}$

d. $\frac{5}{17}$ ✓

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{5}{17} = \frac{5}{17}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

Of 23 students, 13 play soccer and 5 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{1}{2}$

b. $\frac{13}{23}$

c. $\frac{5}{13}$ ✓

d. $\frac{8}{13}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{5}{13} = \frac{5}{13}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Of 30 students, 13 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{13}{30}$

b. $\frac{1}{10}$

c. $\frac{3}{17}$

d. $\frac{3}{13}$ ✓

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{3}{13} = \frac{3}{13}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Of 21 students, 12 play soccer and 7 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{5}{12}$

b. $\frac{7}{9}$

c. $\frac{1}{3}$

d. $\frac{7}{12}$ ✓

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{7}{12} = \frac{7}{12}$.

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

Of 40 students, 28 play soccer and 9 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{19}{28}$

b. $\frac{9}{28}$ ✓

c. $\frac{3}{4}$

d. $\frac{7}{10}$

Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{9}{28} = \frac{9}{28}$.

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